Case Study / Case Report

Management of Mesiodens: A Supernumerary Tooth

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Abstract
Supernumerary tooth is a developmental disturbance associated with the number of teeth. Mesiodens is a supernumerary tooth present in the midline between the two central incisors. It usually results in oral problems such as malocclusion, food impaction, poor aesthetics, and at times cyst formation. This case report deals with the management of a palatally placed, fully erupted mesiodens in the maxillary arch of a 7-year-old male patient, extracted under local anaesthesia following radiographic confirmation of complete root formation.

Key Words: Dental anomalies, Mesiodens, Pediatric dentistry, Supernumerary teeth, Surgical extraction

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I. Introduction

Supernumerary teeth or hyperdontia refers to the presence of excessive teeth than the normal count in the dental arch. The prevalence of supernumerary teeth is 2.2% in the anterior region among which 78.4% represents mesiodens[1]. It is usually found in primary dentition.[2] Mesiodens is often reported with the complications such as spacing, crowding, speech difficulties, resorption of adjacent root, poor esthetics, delayed eruption of permanent incisors, and development of a dentigerous cyst[1]. The anomaly is usually noticed around 7 to 9 years, and most mesiodens are found during the mixed dentition period. Timely removal of mesiodens leads to normal eruption of the central incisors, owing to the eruption path of the lateral incisors and canine.[2] Extraction is generally the preferred treatment even for erupted mesiodens.[3] That is why mesiodens are seldom observed in adults with permanentdentition as most of them get extracted within the first or early second decade of life.[2]

II. Case Report

This report describes the diagnosis and management of a fully erupted mesiodens in a pediatric dental patient.

CASE REPORT

A seven-year-old male patient reported to the Department of Dentistry with the chief complaint of forwardly placed upper anterior teeth and the presence of an extra tooth behind these two teeth. No significant history of presenting illness was revealed. Family history revealed that his mother underwent mesiodens extraction when she was eight years old. Patient also revealed that he was unhappy with the appearance of his teeth. On examination a palatally placed mesiodens was found in relation to 11, 21 (Figure 1). On radiographic examination (Figure 2), a solitarymesiodens with complete root formation was observed. The mesiodens was just pierced the overlying soft tissue mucosa and was erupting into the oral cavity. Extraction ofmesiodens w planned. Treatment plan was discussed wi his father and written informedconsent was taken. The periosteum was elevated using a periosteal elevator, local anesthesia was then administered and the mesiodens was extracted using tooth forceps (Figure 3, 4). The extraction socket was then irrigated thoroughly with 5

III. Discussion

Mesiodens is the commonly encounteredsupernumerary tooth in dental clinics.1Nearly two-thirds of cases of mesiodens occur in males with a mean age of diagnosis being 6.35±1.[85] years.[2] Mostly it is diagnosed during the first decade of life.It'sdevelopment begins during the third trimester ofpregnancy.[1] Mesiodens most often have clinical implications such as delayed eruption, rotation or root resorption of adjacentteeth, malocclusion, and midline diastema. At times, theymay be associated with cystic lesions or even erupt in thenasal cavity.[3] Only less than 20% of cases of mesiodensare free of symptoms without any associated pathology.In the asymptomatic cases of mesiodens, the anomalyis usually diagnosed during routine clinical/radiographic examination.[4],5Yet, in this particular case, the patient did not have any complaints or prior visits to a dental clinic; therefore, the tooth remained unnoticed.

Etiology

There are numerous theories put forwarded byvarious authors to explain the etiology of mesiodens.[6]

Dichotomy theory: Due to certain environmentalfactors like trauma, inflammation, abnormalpressure the tooth germ is divided into two equalor unequal sections resulting in the formation ofsupernumerary teeth.

Genetic theory: Considering the familial pattern,a genetic basis has been suggested for supernumeraryteeth. Nonsyndromic mesiodens is associatedwith the interaction between BMP, SHH, WNT signaling pathways.[7] Decreased expression of USAG and ACVR2A (Activin type II receptor protein) further activates BMP-induced odontogenesis of the supernumerary tooth bud.[8,9,10] Increased WNT and BMP signaling has been reported to produce supernumerary teeth in mice.[11]

Hyperactivity theory: This is the most accepted theory. According to this theory, remnants of the dental lamina develop into an extra tooth bud, resulting in supernumerary tooth.[1]

Associated syndromes: Mesiodens is associatedwith syndromes such as cleidocranial dysplasia (CCD), Gardner's syndrome, cleft palate, Nance-Horan syndrome, Down syndrome, tricho-rhino-phalangeal syndrome, Rothmund-Thomson syndrome.1In the present case, the mesiodens was not associated with any symptoms.

Classification

Mesiodens can be classified based on its shape as Eumorphic or Dysmorphic: conical, tuberculate, molariform, and odontome types.[12]

Recently, a new classification system was given by Albert A and Mupparapu M (2018) for better treatment planning. The classification is based on the angulation of mesiodens to the normal eruptive pattern within the maxilla.[13]

Class 1: Parallel/ 00 to normal eruptive pattern

Class 2: 0-900 to normal eruptive pattern

Class 3: Perpendicular/ 900 to normal eruptive pattern

Class 4: 90-1800 to normal eruptive pattern

Class 5:Inverted/ 1800 to normal eruptive pattern

Mesiodens can be diagnosed by the combination of both clinical and radiographic examination. Abnormal midline diastema in early mixeddentition or proclination of teeth gives a clue to the presence of mesiodens. Radiographic examination shows a tooth-like structure in between the incisors. Generally, Clark's rule (also known as the SLOB technique – same lingual, opposite buccal) is followed to confirm the precise location of the mesiodens.[1]

Different timing for the extraction is suggested in the literature: early intervention and delayed intervention.[14,15] Early intervention, extraction immediately its after diagnosis, is the most preferred. Extraction of the mesiodens in the early mixed dentition period promotes spontaneous eruption and alignment of central incisors. Also, it promotes the early orthodontic intervention there by improving the patient profile at an early age. Late intervention refers to the extraction after the complete root formation of adjacent teeth. This is suggested to avoid potential complications to the developing roots of adjacent teeth.[1]

In the present case the root was completely formed, the crown was easily visible, and the positioning was favourable, so extraction with forceps was planned. The patient was scheduled for three-monthly follow-up for self-correction of the proclination of the upper central incisors, otherwise orthodontic minor correction was planned.

IV. Conclusion

Clinical experience suggests and the dental literature supports that an individualized radiographic examination of any pediatric patient who presents clinical evidence of delayed permanent tooth eruption or temporary tooth displacement with or without a history of previous dental trauma should be performed. Early diagnosis allows adoption of a less complex and less expensive treatment and ensures better prognosis.

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